Session Overview & Concepts

Welcome to the "Girls2Code Student Guides"! We are thrilled to kick off this journey. Remember, every expert was once a beginner, and debugging errors is a universal engineering reality, not a personal failure. Today, you will build your first webpage and set up your professional developer environment.
Before writing code, we need to understand how the internet operates:

Frontend vs. Backend: The frontend is everything you see and interact with (buttons, text, colors). The backend is the hidden logic and database that processes information.

Client and Server: Think of ordering coffee in Malé. The "Client" (your browser) is you ordering the drink. The "Server" is the kitchen preparing and delivering your website.

HTML (HyperText Markup Language): This provides the structural foundation of a webpage using "tags".

Git & GitHub: Git tracks changes to your code, while GitHub is a remote platform to store and collaborate on those projects. A "repository" is simply a project folder that remembers its own history.

The tools that will used in this session are Visual Code (app) and GitHub (Website)
To download Visual Code please go this website:
https://code.visualstudio.com/download?_exp_download=d53503e735_
To Setup GitHub to follow the guide in Page 3.

Knowledge Check: If you are designing the visual layout of a website, are you working on the frontend or backend?

Answer: Frontend

Tools & GitHub Workflow

We will configure tools that mirror professional software engineering environments.

1. **Replit & VS Code:** Replit is a cloud-based development environment utilized for immediate, zero-setup coding, while VS Code is your local professional desktop editor. *We will be using VS Code for now for the ease.*
2. **GitHub Account Setup:** Go to GitHub.com and register. Create your first "repository" (project folder) and include a README.md file to describe it. For more step to step guide follow: Check the Page 3 of the guide.
3. **The Git Habit:** We will build the everyday professional habit of version control: save → test → commit → push.

4. **AI Assistant Note:** Generative AI is a collaborative assistant. You may use it to explain syntax, but you must be able to independently explain any code you add to your project.

Practical Activity: Build Your Webpage

1. **Workspace Setup:** Open Visual Code, create a new HTML project, and locate the index.html file.
2. **Code Along (Structure):** Type <h1>Welcome to My Page</h1> to create a large text heading. Run the code to view the result.
3. **Independent Practice:** Below your heading, add a tag containing a short introduction about yourself.
4. **Adding Media:** Add an image using the <img> tag and create a clickable link using the <a> tag.
5. **Debugging Tip:** If your image is broken, check for typos in the file name. Treat errors as helpful diagnostic clues, not mistakes.

Expected Output & Deliverables

Your completed HTML code should follow this structure:
HTML code is below:

```
<!DOCTYPE html>
<html>
<body>
  <h1>My Tech Journey</h1>
  <p>Hello! I am learning to build digital products.</p>
  <img src="maldives.jpg" alt="A beautiful local island">
  <a href="https://github.com">View my GitHub</a>
</body>
</html>
```

Common Mistake: Forgetting the forward slash on closing tags (e.g., typing instead of). The visual output will be a plain, unstyled document with functional text and media.

- Working HTML webpage with a heading, introduction, image, and link.
- Registered GitHub account.
- First repository successfully created.
- README file containing your participant and project information.

What was the most surprising thing you learned about how web pages are built today?

How to Setup your GitHub account

Step 1: Log In to GitHub Using Your Google Account

1. Open your browser and navigate to the GitHub login page: github.com/login
2. Beneath the standard username/password fields, click the button that says "Continue with Google" (or select the Google icon).
3. A popup/prompt will appear—choose the Google account you want to use.
4. If this is your first time logging in with this Google account:
    - Google will ask you to authorize GitHub. Click Continue / Authorize GitHub.
    - If you don't have an existing GitHub account linked to that email, GitHub will ask you to quickly confirm your desired Username and accept the terms of service.
5. Once authenticated, you will be redirected straight to your GitHub Dashboard.

Step 2: Create a New Repository

1. Once on your GitHub dashboard, locate and click the + (plus icon) in the top-right corner of the page next to your profile picture.
2. Click "New repository" from the drop-down menu. *(Alternatively, click the green "New" button on the left panel)*.
3. Fill in the repository details:
    - Repository name: Type a name for your project (e.g., girls-to-code-2026).
    - Description *(Optional)*: Briefly describe what your project does.
    - Visibility: Choose between:
        - Public: Anyone on the internet can view your repository.
        - Private: Only you (and people you explicitly invite) can view it.
4. Initialize the repository (Recommended):
    - Check "Add a README file" (this provides a space to write documentation for your project).
5. Click the green "Create repository" button at the bottom of the form.

Your repository is now created! You will be taken directly to your new repository page where you can upload files, add code, or share the repository link with others.